Safety-Shielded Candidate-Graph Sequence Improvement for Dynamics-Aware Orbital Inspection

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***Abstract*— Autonomous orbital inspection requires a planner to couple which surface regions are observable with whether consecutive observations are dynamically reachable and auditable. We formulate this problem on a directed candidate graph whose observation nodes store fixed camera-valid masks and whose safe observable orbital arc (SOOA) edges store source-dependent finite-duration Hill–Clohessy–Wiltshire transfers. Visibility and clearance queries use a bounding-volume hierarchy over all 247,525 triangles of the ISS mesh, and sampled transfers are additionally rejected when a motion segment crosses the mesh surface. A shielded sequence-improvement method starts from a feasible weighted-coverage/HCW incumbent, audits every proposed reversal under the same edge model, and returns an alternative only when it reaches the same inspectable-coverage goal with lower graph cost. We prove monotone submodularity of fixed-mask coverage, Markov sufficiency of the finite graph state, preservation of stored per-edge audits under exact rest-to-rest concatenation, and incumbent no-degradation in the shared graph objective. In a predeclared 20-scenario paired study with six-dimensional Latin-hypercube initial-state perturbations, both methods were feasible and reached identical 80.49% inspectable coverage in all scenarios. Sequence improvement reduced Δv in all 20 pairs by a mean 0.232 m/s (95% paired-bootstrap confidence interval: 0.128 m/s to 0.338 m/s; two-sided sign-test $p = 1.91 \times 10^{-6}$), a 1.92% mean reduction. These claims apply to the deterministic sampled graph and do not imply continuous-time safety, robustness to state uncertainty, or continuous-space optimality.**

***Index Terms*— spacecraft inspection, sequence optimization, safe observable orbital arc, relative orbital dynamics, mesh collision checking, coverage planning, Hill–Clohessy–Wiltshire equations**

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I. Introduction

Large orbital structures are increasingly important to crewed flight, science, logistics, and in-space assembly. Their external surfaces must be inspected for configuration changes, impact damage, degraded thermal protection, docking-interface anomalies, and other faults that fixed onboard cameras may not observe. An autonomous inspector must therefore decide both where to look and how to move. These decisions are tightly coupled in orbit because a camera pose with high geometric coverage can be unreachable, control-intensive, or unsafe under relative motion.

Existing planning abstractions often expose only one side of this coupling. Robotic next-best-view methods provide mature treatments of sensor range, occlusion, incidence angle, and marginal coverage, but their motion cost is frequently Euclidean or platform-specific. Relative-orbit design provides dynamically meaningful trajectories, yet surface coverage and camera dwell are often secondary constraints rather than the planning objective. A sequential pipeline that selects views first and checks transfers afterward can therefore overestimate useful coverage. The selected views may be attractive in isolation while the connecting trajectory violates input or keep-out limits.

The deeper issue is the definition of a selectable inspection action. A static viewpoint is incomplete because it contains no evidence that the spacecraft can reach that pose under its relative dynamics. A dynamically natural orbit segment is also incomplete when it does not reveal new weighted surface area. For orbital inspection, the action itself should include the transfer, the stabilized camera observation, and the safety evidence used to admit that action. This requirement motivates a finite representation that binds observability and relative-motion feasibility before the coverage sequence is optimized.

We introduce the safe observable orbital arc (SOOA), a source-dependent finite-duration HCW transfer from the current graph node to a stabilized candidate observation. Each SOOA references its destination's camera-valid target set and stores control effort, minimum sampled clearance, terminal error, input and speed status, and optional passive-drift audit. Geometric viewpoints define graph nodes, but they do not contribute coverage until the corresponding directed SOOA has passed the enabled feasibility tests. Figure 1 summarizes the progression from mesh targets and camera visibility to a directed graph of admissible inspection actions. The graph objective then trades newly covered weighted area against the cost of each reachable arc.

The finite representation makes sequential optimization and verification explicit. Candidate nodes are decision nodes, but a node identifier alone is not Markov because the value of a view depends on what has already been covered and how much budget remains. We therefore augment the graph state with covered- and selected-set masks and apply safety-shielded candidate-graph sequence improvement. A deterministic weighted-coverage/HCW tour supplies the incumbent; bounded reversal search changes its order only after every new directed edge passes the same shield. Exact dynamic programming remains available only for reduced graphs small enough to exhaust.

The contributions are:

1) a node–edge representation in which candidate observations store camera-valid target sets and directed SOOAs store source-dependent HCW motion and sampled audit evidence;
2) a full-mesh bounding-volume hierarchy used consistently for camera line of sight, candidate clearance, sampled trajectory clearance, and swept surface-crossing rejection;
3) a Markov candidate-node formulation and deterministic safety-shielded sequence-improvement algorithm based on audited reversal moves and a feasible incumbent safeguard;
4) proofs of Markov sufficiency, compositional preservation of stored audits, graph-objective incumbent no-degradation, and exact optimality on exhausted reduced graphs; and
5) a predeclared 20-scenario paired experiment with full result provenance, paired bootstrap confidence intervals, and an exact sign test.

II. Related Work

Relative-motion models provide the dynamic foundation for orbital inspection. The Clohessy–Wiltshire equations remain a common baseline near a circular chief orbit because they are linear, interpretable, and inexpensive to propagate [1], [2]. Higher-fidelity flight design must account for nonlinear gravity, perturbations, navigation error, and actuator dynamics, but HCW propagation is useful for isolating how view ordering changes control effort and safety under a fixed model. Visibility-constrained orbit design has also been studied directly. Hou et al. analyze relative orbits for visual inspection of the China Space Station [3], while Zhou et al. combine near-natural relative ellipses with formation control around a rotating station model [4]. These approaches provide structured orbital motion and stronger closed-loop control analysis than the present planner, but their primary object is an orbit or formation rather than a sequence of surface-coverage actions.

Robotic inspection provides the complementary view-selection perspective. Classical next-best-view work models sensor range, incidence, occlusion, reconstruction quality, and motion cost [5]. Sampling-based kinodynamic planners such as RRT and asymptotically optimal variants incorporate differential constraints and collision avoidance in continuous state space [6], [7]. These methods are valuable when online reachability or continuous obstacle avoidance dominates. However, surface inspection additionally requires persistent accounting of which weighted targets have satisfied visibility and dwell. That reward must be attached to the motion primitive or repeatedly recomputed during search.

Approximate dynamic programming and rollout address sequential decisions when exact Bellman recursion is too large [8], [9]. Learning-based policies nevertheless require a separate safety argument: constrained policy optimization controls expected constraint costs [10], whereas shielding removes actions that violate an independently specified safety model [11]. The present contribution uses the latter separation but does not require a learned value function. Every local sequence proposal is evaluated by the model-based HCW, input, mesh-clearance, terminal, and optional passive-drift audits before it can replace the incumbent.

Recent spacecraft-inspection studies increasingly couple geometry and motion. Apodaca et al. address inspection trajectory generation and coverage–energy trade-offs for in-orbit structures [12]. Zhou et al. study surface-aware tomographic paths around complex spacecraft geometry [13]. Ogri et al. address an execution-level problem involving image-feature tracking, information gain, and dwell under switched-system logic [14]. Together, these studies show that geometric coverage, orbital dynamics, and online sensing cannot be treated as fully independent. They also differ in the level at which the coupling occurs: orbit design, continuous path generation, online camera control, or discrete inspection planning.

OrbInspect is positioned at the discrete-action level. Its contribution is neither a new relative dynamics model nor a flight-certified collision-avoidance controller. Each SOOA is simultaneously a sensor-valid coverage contributor and a dynamically audited finite motion, so an infeasible transfer cannot improve the planner's reported coverage. This representation gives more detailed camera and sequential view-selection logic than formation-level observation studies, while those studies provide lessons for future work: near-natural relative arcs can seed lower-effort SOOA libraries, and a rotating body frame should replace the fixed-LVLH target assumption when station attitude motion is material. The current optimality claim remains conditional on the sampled library, and continuous-space global optimality is outside scope.

III. System and Observation Model

The present study addresses offline planning. It receives a known station mesh, an initial relative state, camera parameters, a keep-out model, and actuation limits, and returns a time-indexed translational reference, a camera-boresight stream, selected SOOA records, and summary metrics. A future ROS 2 replay layer may publish these records, but no ROS or Gazebo result is used to support the quantitative claims in this manuscript.

A. Relative Dynamics

The inspected structure is fixed in an LVLH frame attached to a nominal circular chief orbit. The axes follow

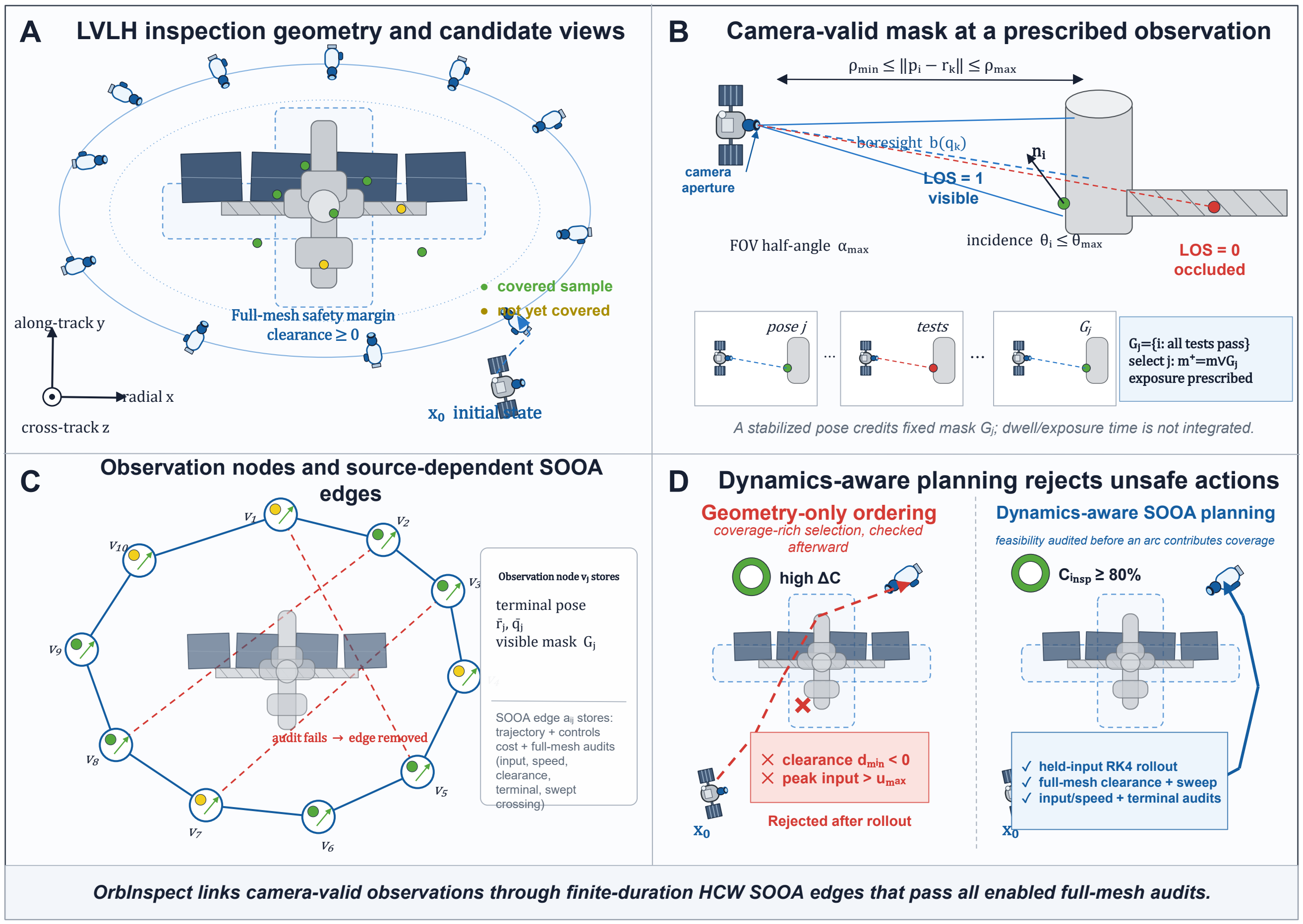

Fig. 1. OrbInspect formulation. (a) Candidate observations are generated around an ISS-scale structure and outside a mesh-derived safety margin. (b) Range, field of view, incidence, full-mesh line of sight, and the prescribed observation condition determine each node's visible-target set. (c) A directed SOOA edge binds a source-dependent HCW transfer and its sampled and swept-mesh audits to a destination observation. (d) Sequence improvement excludes coverage-rich choices whose incoming edge violates an enabled audit.

the radial, along-track, and cross-track directions shown in Fig. 1. The chaser state is $\mathbf{x} = [\mathbf{r}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^6$, where $\mathbf{r} = [r_x, r_y, r_z]^\top$ and $\mathbf{v} = [v_x, v_y, v_z]^\top$. With mean motion n and commanded acceleration $\mathbf{u} = [a_x, a_y, a_z]^\top$, the HCW model is

$$
\begin{aligned}
&\dot{\mathbf{x}} = A_{\mathrm{HCW}}\mathbf{x} + B\mathbf{u},\\
&A_{\mathrm{HCW}} = \begin{bmatrix} 0 & I \\ K_n & C_n \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ I \end{bmatrix},\\
&K_n = \mathrm{diag}(3n^2, 0, -n^2), \quad C_n = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.
\end{aligned} \tag{1}
$$

For the continuous HCW model, piecewise-constant acceleration over Δt has the exact zero-order-hold map

$$
\begin{aligned}
\mathbf{x}_{k+1} &= A_d\mathbf{x}_k + B_d\mathbf{u}_k,\\
A_d &= e^{A_{\mathrm{HCW}}\Delta t},\\
B_d &= \int_0^{\Delta t} e^{A_{\mathrm{HCW}}\tau} B \, d\tau.
\end{aligned} \tag{2}
$$

The reported implementation uses the same piecewise-constant input model but constructs and propagates each discrete boundary map with fixed-step fourth-order Runge–Kutta integration. We denote that numerical affine map by $\mathbf{x}_{k+1} = \widetilde{A}_d\mathbf{x}_k + \widetilde{B}_d\mathbf{u}_k$; Eq. (2) is the exact continuous-model reference, not a claim that the archived samples were generated by a matrix exponential. The transfer is therefore a finite-duration continuous-thrust maneuver with zero-order-held acceleration, which may jump at sample boundaries, rather than an impulsive maneuver. Each segment is solved rest-to-rest at a candidate observation, and the numerical rollout is accepted only when its terminal position and velocity errors satisfy the stated tolerances.

B. Visibility, Prescribed Observation, and Coverage

The structure is represented by weighted surface samples $\mathcal{T} = \{(p_i, n_i, w_i)\}_{i=1}^{N}$ with $w_i > 0$. Each weight approximates the surface area represented by its sample, so a densely tessellated component does not dominate solely because it contains more vertices. The camera attitude q_k defines unit boresight $b(q_k)$, and $\mathcal{O}$ denotes

TABLE I
Modeling assumptions and the boundary of the reported evidence.

Assumption	Role in the baseline	Consequence for interpretation
Circular chief orbit	Supports linear HCW relative translation in the LVLH frame.	Quantitative results compare planners under the same local linear model, not a perturbed flight ephemeris.
Station fixed in LVLH	Keeps mesh targets and the mesh safety geometry stationary during one planning run.	Rotating appendages or station attitude motion require a body-frame extension.
Known mesh and state	Makes visibility, occlusion, and transfer evaluation deterministic.	Navigation, map, and shape uncertainty are not represented by the present safety margin.
Prescribed camera attitude	Points the boresight toward each candidate observation target and exports the corresponding quaternion stream.	Camera slew time, reaction-wheel limits, and closed-loop attitude error are outside the optimization.
Mesh-based sampled safety	Computes distance to the full ISS triangle mesh at every rollout sample and rejects sampled motion segments that intersect the mesh.	Positive sampled clearance and surface nonintersection are planning audits rather than a robust continuous-time collision certificate.
No illumination or plume model	Isolates view geometry and translational feasibility.	Sun angle, shadow, contamination, and plume-impingement restrictions must be added for flight design.

the occluding mesh. Target i is visible when

$$\begin{aligned} r_{\min}^{\mathrm{cam}} &\leq \|p_i - \mathbf{r}_k\| \leq r_{\max}^{\mathrm{cam}}, \\ \arccos\frac{(p_i-\mathbf{r}_k)^\top b(q_k)}{\|p_i-\mathbf{r}_k\|} &\leq \alpha_{\max}, \\ \arccos\frac{(\mathbf{r}_k-p_i)^\top n_i}{\|\mathbf{r}_k-p_i\|\,\|n_i\|} &\leq \theta_{\max}, \\ \mathrm{LOS}(\mathbf{r}_k, p_i, \mathcal{O}) &= 1. \end{aligned} \tag{3}$$

The four tests exclude targets that are too near or too far, outside the field of view, observed at an excessive incidence angle, or blocked by station geometry. Line of sight is queried against a deterministic bounding-volume hierarchy containing every ISS mesh triangle; intersections at the target endpoint are excluded, while interior segment intersections block the view. In a closed-loop execution model, dwell could be accumulated as $\tau_{i,k+1} = \tau_{i,k} + \Delta t$ while target i remains visible and accepted when $\tau_i \geq \tau_{\min}$. The offline study instead treats every candidate as a stabilized terminal observation with its prescribed exposure completed a priori. Selecting node j therefore credits its fixed visibility mask G_j, so $m_{k+1} = m_k \vee G_j$. The weighted coverage represented by mask m_k is

$$C(m_k) = \frac{\sum_i w_i m_{i,k}}{\sum_i w_i}. \tag{4}$$

Candidate generation can make a subset of the sampled surface invisible from every candidate observation. For the nonempty set $\mathcal{T}_{\mathrm{insp}} = \cup_{j=1}^{M} G_j$, we therefore also report inspectable coverage:

$$C_k^{\mathrm{insp}} = \frac{\sum_{i\in\mathcal{T}_{\mathrm{insp}}} w_i m_{i,k}}{\sum_{i\in\mathcal{T}_{\mathrm{insp}}} w_i}. \tag{5}$$

The total coverage in Eq. (4) measures how much of the sampled structure is credited, whereas Eq. (5) conditions on the candidate library. The reported mission duration includes transfer time but no additional dwell or exposure time. Attitude determines the visibility predicate and exported boresight; closed-loop attitude dynamics, illumination, and interruption of a dwell are outside the offline model.

C. Dynamic Feasibility and SOOAs

Every sampled state must satisfy

$$\|\mathbf{u}_k\|_2 \leq u_{\max}, \qquad \|\mathbf{v}_k\|_2 \leq v_{\max}, \qquad d_{\mathcal{S}}(\mathbf{r}_k) \geq 0, \tag{6}$$

where $d_{\mathcal{S}}(\mathbf{r}) = d_{\mathrm{mesh}}(\mathbf{r}) - d_{\mathrm{safe}}$ is clearance above the required margin and d_{mesh} is unsigned distance to the closest triangle in the complete ISS mesh; thus $d_{\mathcal{S}} = 0$ is the sampled admissibility boundary. In addition, each consecutive pair of rollout samples is tested for a triangle-surface intersection. This swept test catches a direct mesh crossing even if both endpoints have positive unsigned distance. For a directed rollout from source node i to destination node j, define $\Delta v_{ij} = \sum_k \|\mathbf{u}_{ij,k}\|_2 \Delta t$, root-mean-square distance e_{ij}^{rms} from the transfer samples to the destination position, and $d_{ij}^{\min} = \min_k d_{\mathcal{S}}(\mathbf{r}_{ij,k})$. The graph stage cost used in the reported study is

$$\ell_{ij} = \lambda_v \Delta v_{ij} + \lambda_e e_{ij}^{\mathrm{rms}} + \lambda_s \max(0, -d_{ij}^{\min}) + \lambda_a. \tag{7}$$

For every shield-admissible edge, the clearance-violation term is zero. It remains in the shared cost implementation so deliberately unshielded comparison methods can be evaluated with the same expression.

A candidate observation node and a directed SOOA edge are, respectively,

$$\begin{aligned} v_j &= (\bar{\mathbf{r}}_j, \bar{q}_j, G_j), \\ a_{ij} &= \big(\mathbf{x}_{ij}[1{:}K], \mathbf{u}_{ij}[0{:}K-1], q_{ij}[1{:}K], \\ &\qquad \Delta v_{ij}, d_{ij}^{\min}, u_{ij}^{\max}, v_{ij}^{\max}, e_{ij}^{r}, e_{ij}^{v}, \rho_{ij}\big). \end{aligned} \tag{8}$$

Here $\bar{\mathbf{r}}_j$ and $\bar{q}_j$ are the terminal observation pose and prescribed attitude, $G_j \subseteq \mathcal{T}$ is its camera-valid target set, and e_{ij}^{r} and e_{ij}^{v} are terminal position and velocity errors. The stored state and attitude arrays contain the K post-step samples; the source state is held by node v_i. The initial condition is a distinguished source node v_0 with no visibility reward. For passive horizon $T_p = L_p \Delta t$, an optional audit can propagate zero-input HCW drift from every stored edge state and record $\rho_{ij} = \min_{1\leq k\leq K,\, 0\leq l\leq L_p} d_{\mathcal{S}}(\mathbf{r}_{ij,k}^{\mathrm{drift}}(l))$. The paired study disables this optional audit and therefore makes no passive-safety claim; it retains input, speed, full-mesh sampled clearance, swept surface-intersection, and terminal checks. Throughout this paper, "safe" means admissible under the audits explicitly enabled for the reported experiment. These deterministic checks do not certify robustness between samples or under model and state uncertainty.

IV. Safety-Shielded Candidate-Graph Sequence Improvement

A. Candidate Graph and Markov State

Let $\mathcal{V} = \{v_0, v_1, \ldots, v_M\}$ contain the initial node and the M candidate observation nodes in Eq. (8). A directed edge from v_i to v_j is the source-dependent SOOA a_{ij}, and it is absent when an enabled audit fails. Thus visibility G_j is node dependent, whereas trajectory, cost, clearance, input, terminal error, and passive margin are edge dependent. This separation is necessary because reaching the same observation from two different source nodes generally produces different HCW motion and audit values.

A node identifier is not a sufficient decision state because the value of a view changes after its targets have been covered. The finite-horizon Markov state is

$$s_k = (j_k, m_k, b_k, h_k), \tag{9}$$

where $j_k \in \{0, \ldots, M\}$ is the current node, $m_k \in \{0,1\}^N$ is the covered-target mask, $b_k \in \{0,1\}^M$ is the selected-observation mask, and h_k is the remaining action budget. Selecting destination $a \in \{1, \ldots, M\}$ traverses edge $a_{j_k a}$ and gives the deterministic transition

$$f(s_k, a) = (a,\ m_k \vee G_a,\ b_k \vee \mathbf{e}_a,\ h_k - 1)\,. \tag{10}$$

The shield-admissible action set $\mathcal{U}_s(s)$ is defined below. The constrained graph problem assigns zero terminal cost after reaching C_{req} and value $+\infty$ when no goal-reaching continuation exists within the remaining budget. Its Bellman recursion is

$$\begin{aligned} V^\star(s) &= \begin{cases} 0, & C(m) \geq C_{\text{req}}, \\ +\infty, & h = 0 \text{ or } \mathcal{U}_s(s) = \varnothing, \\ \min\limits_{a \in \mathcal{U}_s(s)} Q^\star(s,a), & \text{otherwise}, \end{cases} \\ Q^\star(s,a) &= \ell_{ja} + V^\star(f(s,a)). \end{aligned} \tag{11}$$

Here j and h are the current-node and remaining-budget components of s. The infinite value enforces the coverage constraint in the exact oracle; the reversal method instead compares only complete goal-reaching sequences.

B. SOOA Construction and Safety Shield

Area-weighted targets are sampled from the mesh, and camera seeds are generated on offset shells around their outward normals. The binary visibility matrix $V_{ij} = \mathbf{1}\{i \in G_j\}$ is computed once from the terminal observation geometry. For a transition to $\bar{\mathbf{x}}_j = [\bar{\mathbf{r}}_j^\mathsf{T}, \mathbf{0}^\mathsf{T}]^\mathsf{T}$, the planner solves the minimum-energy piecewise-constant HCW boundary-value problem under the numerical RK4 map

$$\begin{aligned} \min_{\mathbf{u}_0,\ldots,\mathbf{u}_{K-1}} \quad & \sum_{k=0}^{K-1} \|\mathbf{u}_k\|_2^2 \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = \widetilde{A}_d \mathbf{x}_k + \widetilde{B}_d \mathbf{u}_k, \\ & \mathbf{x}_K = \bar{\mathbf{x}}_j. \end{aligned} \tag{12}$$

If the terminal influence matrix $\mathcal{B}_K = [\widetilde{A}_d^{K-1}\widetilde{B}_d \ \cdots \ \widetilde{B}_d]$ has full row rank, the stacked minimum-norm command from source state $\mathbf{x}_i^{\text{src}}$ is $\mathbf{U}_{ij}^\star = \mathcal{B}_K^\mathsf{T}(\mathcal{B}_K\mathcal{B}_K^\mathsf{T})^{-1}(\bar{\mathbf{x}}_j - \widetilde{A}_d^K \mathbf{x}_i^{\text{src}})$. The implementation computes this expression through the 6×6 Gram system and then propagates the requested input without clipping. Let $\chi_{ij} = 1$ exactly when every stored edge sample satisfies Eq. (6), $e_{ij}^r \leq \varepsilon_r$, $e_{ij}^v \leq \varepsilon_v$, and $\rho_{ij} \geq 0$ when the passive audit is enabled. The shield then admits

$$\mathcal{U}_s(s) = \left\{ a \in \{1, \ldots, M\} : \begin{array}{l} b_a = 0,\ G_a \setminus m \neq \varnothing, \\ \chi_{ja} = 1 \end{array} \right\}, \tag{13}$$

where j is the current-node component of s. Unlike a reward penalty, the shield makes an inadmissible edge unavailable to incumbent construction, local search, and exact recursion [11]. The guarantee remains conditional on the sampled numerical rollout and the deterministic model used to compute χ_{ij}.

C. Feasible Incumbent and Audited Reversal Search

Exact recursion is exponential in the coverage and selected-node masks, so the reported method improves a deterministic incumbent rather than approximating the full Bellman value. The incumbent first applies weighted set cover over the inspectable target set and then orders the selected observations by feasible HCW dynamic cost. Its fixed sequence is replayed through the shield; if any required edge fails or the sequence does not reach $C_{\text{req}}^{\text{insp}}$ within H actions, it is not a valid incumbent.

Starting from a valid sequence $(j_1, \ldots, j_L)$, one local move reverses a contiguous subsequence $j_p, \ldots, j_q$. The proposal is evaluated from the initial node with an empty coverage mask. Every source-dependent edge is recomputed or loaded from the same deterministic cache, shielded under Eq. (13), and accumulated until the inspectable-coverage goal is first reached. Consequently, a reversal can reduce cost both by replacing directed transitions and by reaching the goal before trailing observations are needed. Among all successful reversals in one pass, the method accepts the lowest-cost strict improvement. It repeats this best-improvement pass up to a configured limit and returns the lower-cost of the final sequence and the unchanged incumbent.

D. Properties and Exact Oracle

PROPOSITION 1 (Monotone submodularity of fixed-mask coverage) *For an observation set $S \subseteq \{1, \ldots, M\}$, let*

Algorithm 1 Safety-shielded candidate-graph sequence improvement

Require: Candidate nodes, lazy directed-SOOA evaluator, goal C_{req}, budget H

```
Build and audit the deterministic incumbent μ_0
if μ_0 does not reach C_req through shielded edges then
    return infeasible
end if
π̂ ← μ_0
for at most P best-improvement passes do
    π_best ← π̂
    for each contiguous subsequence with at least two nodes do
        Reverse the subsequence and replay it through the shield
        if the proposal reaches C_req within H and lowers J then
            π_best ← proposal if it is cheapest in this pass
        end if
    end for
    if J(π_best) ≥ J(π̂) then
        break
    end if
    π̂ ← π_best
end for
return arg min{J(μ_0), J(π̂)} and all audit records
```

$F(S) = \sum_{i=1}^{N} w_i \mathbf{1}\{i \in \cup_{j\in S} G_j\}$. *If all* $w_i \geq 0$, *then* F *is normalized, monotone, and submodular.*

Proof:
Clearly $F(\varnothing) = 0$. If $A \subseteq B$, every target covered by A is covered by B, so nonnegative weights give $F(A) \leq F(B)$. For $a \notin B$, the marginal gain $F(A \cup \{a\}) - F(A)$ is the total weight in $G_a \setminus \cup_{j\in A} G_j$. Since $A \subseteq B$, this set contains $G_a \setminus \cup_{j\in B} G_j$, proving diminishing returns. ■

Remark 1 (Role of submodularity) Proposition 1 justifies marginal-coverage screening at the node level. It does not supply a greedy approximation ratio for the complete inspection tour because the directed SOOA cost and admissibility of the next observation depend on the current node.

PROPOSITION 2 (Markov sufficiency) *For a fixed deterministic candidate graph with fixed node masks, source-dependent edge records, target weights, revisit rule, and horizon, the state in Eq.* (9) *is Markov for the finite graph decision problem.*

Proof:
The current node j_k identifies the source of every possible next edge; m_k determines marginal and terminal coverage; b_k determines which observations remain selectable under the no-revisit rule; and h_k determines the remaining budget. By assumption, visibility masks and edge costs and audits are deterministic records indexed only by the current and destination nodes. Equations (10), (11), and (13) therefore depend on no earlier history once (j_k, m_k, b_k, h_k) is given. The proposition concerns this finite graph abstraction, not the underlying uncertain continuous spacecraft system. ■

PROPOSITION 3 (Compositional preservation of stored audits) *Consider a sequence* $(a_{j_{k-1}j_k})_{k=1}^{L}$ *whose first source is* v_0, *whose idealized rest-to-rest endpoints match exactly, and whose every edge has* $\chi_{j_{k-1}j_k} = 1$. *Then every stored sample of the concatenated sequence satisfies the input, speed, and clearance inequalities, every edge endpoint satisfies its terminal conditions, and every stored passive-drift rollout has nonnegative margin when that audit is enabled.*

Proof:
By the definition of χ_{ij}, all stated conditions hold on each accepted edge record. Exact endpoint matching identifies the terminal state of edge k with the source state of edge $k+1$, so the records concatenate without an unmodelled jump. The sample set of the complete trajectory is the finite union of the per-edge sample sets; therefore every member of that union retains the corresponding per-edge audit result. ■

Remark 2 (Meaning of audit preservation) The proposition is a finite-record composition result, not a continuous-time barrier certificate. It does not cover inter-sample clearance, endpoint perturbations beyond tolerance, closed-loop tracking error, or robustness to model and state uncertainty.

THEOREM 1 (Incumbent no-degradation in the graph objective) *Let* Π_{cand} *be the finite set compared in the final line of Algorithm 1, and let* $\Pi_F \subseteq \Pi_{\text{cand}}$ *contain exactly the sequences that reach* C_{req} *within* H *actions using only edges in Eq.* (13). *Define* $J(\pi) = \sum_{a_{ij}\in\pi} \ell_{ij}$. *If the incumbent* $\mu_0 \in \Pi_F$ *and the algorithm returns* $\widehat{\pi} \in \arg\min_{\pi\in\Pi_F} J(\pi)$, *then* $\widehat{\pi}$ *is graph feasible and* $J(\widehat{\pi}) \leq J(\mu_0)$.

Proof:
The assumption $\mu_0 \in \Pi_F$ makes the feasible comparison set nonempty. Since $\widehat{\pi}$ minimizes J over a set that contains μ_0, $J(\widehat{\pi}) \leq J(\mu_0)$, and membership in Π_F supplies the stated graph feasibility. ■

Remark 3 (Scope of the safeguard) No-degradation concerns the common stage-cost sum and requires a feasible incumbent in the comparison set. It does not imply lower Δv, shorter runtime, or improvement of any individual metric unless that conclusion is established separately.

For reduced-graph certification, exact recursion caches Eq. (11) for every reachable tuple (j, m, b, h) and enumerates every action in $\mathcal{U}_s(s)$ rather than a local reversal neighborhood.

THEOREM 2 (Optimality on an exhausted reduced SOOA graph) *For fixed finite nodes, deterministic visibility masks, deterministic audited edge costs, a coverage* *requirement, and horizon H, exhaustive evaluation of Eq. (11) returns a minimum-cost goal-reaching shield-feasible sequence when one exists. If no such sequence exists, it returns value $+\infty$ and reports infeasibility.*

Proof:
Proceed by induction on the remaining budget h. At $h = 0$, Eq. (11) returns zero exactly for a goal state and $+\infty$ otherwise, which is correct. Assume the recursion is correct for all states with at most $h - 1$ remaining actions. At a state with h actions, every feasible sequence begins with exactly one $a \in \mathcal{U}_s(s)$ and then a feasible continuation from $f(s, a)$; by the induction hypothesis, the recursion assigns that continuation its minimum cost or $+\infty$ if none exists. Minimizing over every admissible first action therefore returns the minimum total cost, or $+\infty$ when all continuations are infeasible. The claim is limited to the exhausted sampled graph and does not extend to omitted camera poses or continuous-space trajectories. ■

Remark 4 (Graph certificate versus mission optimality) Theorem 2 certifies only the nodes, edges, masks, and audits included in the exhausted graph. Omitted camera poses, alternative transfer durations, and continuous-space trajectories remain outside the certified decision set.

V. Experimental Protocol

A. Paired ISS-Mesh Benchmark

The benchmark uses the public NASA ISS glTF mesh [15]. The transformed mesh contains 247,525 triangles and is sampled into 90 equal-represented-area targets. Two offset shells at nominal radii of 34 and 42 m, with stride two over target seeds, yield 83 safe camera candidates; 41 targets are visible from at least one candidate. The sequence optimizer uses a coverage-preserving pool of at most 48 nodes and a 20-action budget. Every rest-to-rest segment lasts 90 s and uses fixed-step RK4 propagation with $\Delta t = 3\ s$. The HCW mean motion is 1.13137×10^{-3} rad/s, $u_{\max} = 0.060$ m/s^2, $v_{\max} = 2.0\ m/s$, $d_{\text{safe}} = 2\ m$, $\varepsilon_r = 0.5\ m$, and $\varepsilon_v = 0.05\ m/s$. Equation (7) uses $(\lambda_v, \lambda_e, \lambda_s, \lambda_a) = (5, 0.18, 120, 0.05)$.

Both algorithms stop when they first reach $C^{\text{insp}} \geq 0.80$. This common stopping rule prevents an unreachable mesh sample from making the graph goal impossible and prevents cost comparisons at different prescribed goals. Feasibility requires every selected edge to satisfy input, speed, full-mesh sampled clearance, swept mesh nonintersection, and terminal checks. The passive-drift audit is disabled in this paired experiment to keep the claim specific to the implemented mesh-consistent motion audit; passive safety is not inferred from these data. Targets, candidates, the full-mesh visibility matrix, and candidate-to-candidate transfer records are shared across paired scenarios. Planning time is therefore treated as an implementation diagnostic, not a cold-cache algorithm comparison.

B. Predeclared Paired Statistical Design

The study fixes 20 initial states before either method is evaluated. A deterministic Latin-hypercube design perturbs all six components around $[0, -35, 10, 0, 0, 0]^{\mathsf{T}}$: position half-widths are $[8, 6, 6]$ m and velocity half-widths are $[0.012, 0.012, 0.012]$ m/s. Scenario generation uses seed 20260807. Both methods receive the same initial state in each pair and otherwise share the fixed mesh, targets, candidate library, and numerical settings.

The primary endpoint is the paired graph-cost difference $D_J = J_{\text{local}} - J_{\text{inc}}$ among scenarios in which both methods are feasible and reach the inspectable-coverage goal. Secondary endpoints are the paired Δv difference, inspectable-coverage difference, success counts, and minimum mesh clearance. We report the mean and median paired differences, a percentile confidence interval for the mean from 10,000 paired bootstrap resamples, and an exact two-sided sign test after excluding numerical ties. All success counts and coverage differences are reported so cost reduction cannot be interpreted apart from goal attainment.

C. Saved Results and Reproducibility Boundary

The study writes the predeclared initial states, one row per method and scenario, one paired-difference row per scenario, machine-readable summary statistics, and a configuration snapshot. Its result manifest records the result and geometry-query schema versions, Git commit and dirty-worktree flag, configuration SHA-256, ISS mesh SHA-256 and size, Python version, and platform. The case-study generator additionally saves its outcome-independent selection rule, reconstructed trajectory and viewpoint streams, action-indexed coverage and effort, and a figure-to-data hash manifest; it aborts if the reconstructed sequences or primary metrics differ from the paired archive. The current code also validates a 26-feature dimension for optional critic checkpoints, but no critic is enabled or used as evidence in this study. A legacy ten-weight archive is retained only with an explicit incompatibility notice and is not used by the revised manuscript.

D. ROS 2 Execution Interface and Validation Placeholder

The offline records connect to a ROS 2 Jazzy execution stack without making Gazebo physics the source of truth. The trajectory replay node reads the saved state, viewpoint, and attitude streams and publishes position, state, and attitude references on dedicated chaser topics. A tracking controller produces a nominal acceleration, the safety filter publishes the command consumed by the HCW dynamics node, and the dynamics node publishes chaser odometry. The evaluation node records trajectory, control, coverage, safety, planner, and mission-event CSV files together with a rosbag; RViz2 and Gazebo Harmonic

TABLE II

Paired methods evaluated under identical geometry, dynamics, shield, and stopping rules.

Method	Selection rule	Purpose in the comparison
Shielded sequence improvement	Algorithm 1 with four best-improvement reversal passes and final incumbent comparison.	Tests whether audited reordering lowers the shared graph objective without changing the goal or shield.
Shielded two-stage incumbent	Construct a weighted-coverage subset and order it by feasible HCW dynamic cost, then replay it through the same shield.	Supplies the paired base sequence and no-degradation reference.

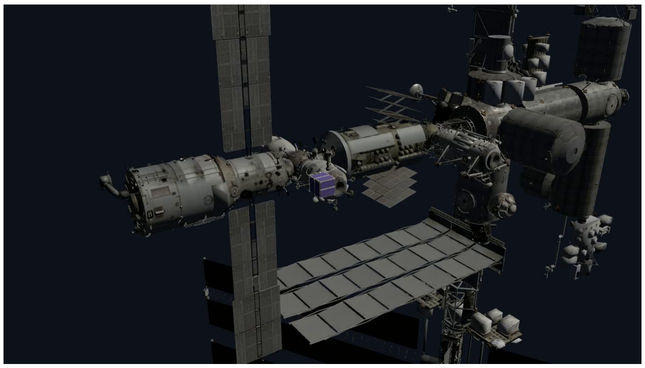

Fig. 2. ROS 2/Gazebo integration placeholder based on an archived infrastructure smoke frame. The ISS and chaser rendering confirms model loading and visualization bridging only; it predates the proposed graph policy and is not used as performance evidence. Before submission, this frame should be replaced by the Jazzy closed-loop SOOA replay with planned and executed paths and the corresponding tracking and safety records.

visualize the same ROS state. This interface preserves the intended separation between planning, closed-loop execution, state propagation, logging, and visualization.

The baseline Jazzy workspace previously completed a short infrastructure smoke run, but that run predates the current SOOA graph planner and therefore does not validate the proposed policy. The required submission experiment is a closed-loop replay of the archived primary route on Ubuntu 24.04 with ROS 2 Jazzy. It must report planned-versus-executed position error, realized Δv, minimum executed clearance, safety-filter intervention count, achieved coverage, dropped-reference count, and completion status. These values are intentionally left unreported until the target-platform run is completed; no synthetic ROS measurements are used in the simulation comparisons.

VI. Results

A. Paired Success, Coverage, and Safety

Both methods were feasible and reached the prescribed inspectable-coverage goal in all 20 paired scenarios. They stopped at the same 80.49% inspectable coverage in every pair, so the effort comparison is not confounded by different credited coverage. Table III reports aggregates over the common-success set, which is the complete scenario set here. The local method used 10.0 SOOAs on average, compared with 11.4 for the incumbent, and reduced mean Δv from 12.085 to 11.853 m/s; Fig. 3(c) resolves the three observed action-count transitions. Every selected transfer stayed below the $0.060\ \mathrm{m/s^2}$ input limit. The worst clearance above the 2 m mesh margin was 0.0619 m for both methods, and no swept segment intersected the mesh. Figure 4 reports the recorded clearance and input audits scenario by scenario.

B. Paired Effect Estimates

Table IV gives the predeclared paired endpoints, defined as local minus incumbent, while Fig. 3(a)–(b) retains every raw paired outcome. Sequence improvement lowered both graph cost and Δv in all 20 scenarios. The mean Δv difference was -0.232 m/s, with a 95% paired-bootstrap confidence interval from -0.338 m/s to -0.128 m/s; relative to the incumbent mean, this is a 1.92% reduction. The exact two-sided sign-test value is 1.91×10^{-6}. The graph-cost result is consistent: the mean difference is -1.911 with a 95% interval $[-2.819, -1.060]$, also with 20 improvements and no losses or ties. Inspectable coverage is tied by construction and in observation: its paired difference is zero in all scenarios.

C. Interpretation and Evidence Boundary

The result supports a specific claim: within the fixed mesh-derived candidate graph, a small number of fully re-audited reversal passes consistently improves the feasible two-stage sequence for the sampled initial-state region. The improvement is modest in relative terms but is paired, coverage matched, and observed in every scenario. It does not show that reversal search is globally optimal, that the candidate library represents all useful observations, or that the same effect holds under a different mesh, transfer duration, or uncertainty model. The Latin-hypercube states are a designed stress set rather than samples from an operational probability distribution.

Full-mesh consistency is material to that interpretation. The current result never substitutes a differently oriented station-primitive proxy for ISS clearance and never estimates occlusion from a stride sample of triangles. The bounding-volume hierarchy queries the complete transformed mesh for both visibility and distance, while swept segment tests reject surface crossings. These corrections invalidate direct comparison with earlier archived numbers generated under the legacy geometry path; the

TABLE III
Aggregate paired-study results over 20 common successful scenarios. Clearance is measured above the required 2 m distance to the full ISS mesh.

Method	Success	C^{insp} (%)	Mean SOOAs	Mean Δv (m/s)	Mean J	Worst d_S (m)
Shielded sequence improvement	20/20	80.49	10.0	11.853	97.289	0.0619
Shielded two-stage incumbent	20/20	80.49	11.4	12.085	99.200	0.0619

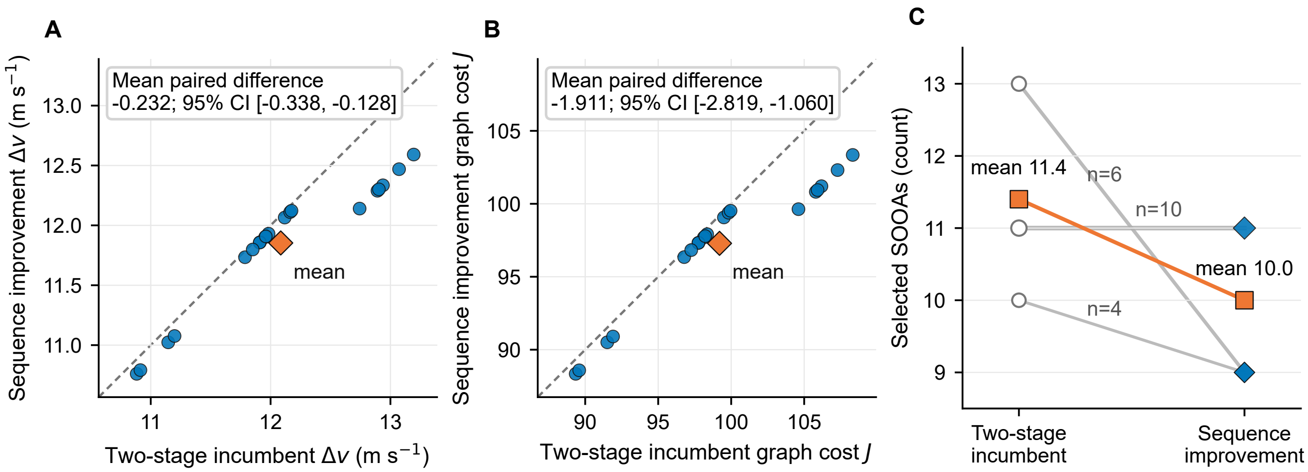

Fig. 3. Paired performance for the 20 predeclared initial states. (a) Total Δv and (b) graph cost; dashed lines denote parity, blue circles denote scenarios, and orange diamonds denote means. All points lie below parity; annotations give the mean local-minus-incumbent difference and 95% paired-bootstrap confidence interval. (c) Action counts; n labels give scenario multiplicities and orange squares give means. Both methods attain common success and identical credited coverage in every pair.

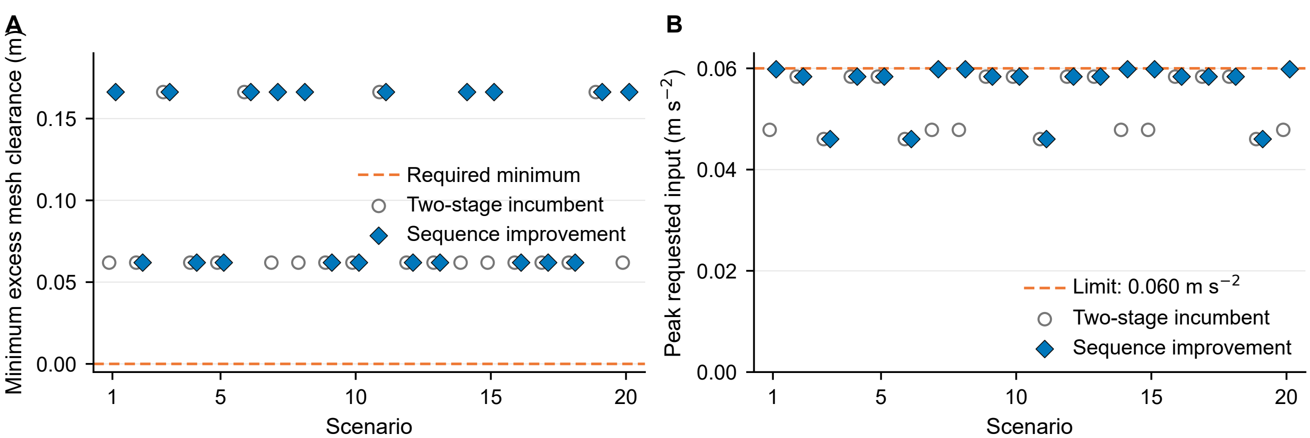

Fig. 4. Recorded safety diagnostics by scenario. (a) Minimum sampled clearance above the required 2 m distance to the full ISS mesh; the dashed zero line is the acceptance threshold. (b) Peak requested acceleration; the dashed line is the configured $0.060\ \mathrm{m/s^2}$ limit. Every value remains admissible. Passive drift, navigation error, and actuator uncertainty are not assessed.

revised paper therefore does not use those numbers as evidence.

D. Outcome-Independent Trajectory Case Study

Aggregate paired effects do not show how sequence improvement changes the executed route. To expose that mechanism without selecting a favorable outcome, we chose the Latin-hypercube state nearest the nominal initial condition after normalizing each of the six coordinates by its predeclared sampling half-width. This design-only rule selects case 9, with normalized distance 0.758 and

$$\mathbf{x}_0 = [2.8, -36.5, 7.3, 0.003, 0.0042, -0.0006]^{\mathsf{T}}. \quad (14)$$

The selection rule does not use either method's cost, coverage, feasibility, or action count. The reconstruction reproduces the archived candidate sequences and paired metrics exactly.

Figure 5 shows every saved HCW position sample and terminal observation pose. Both methods use the same 11 observations. The improved route exchanges the order

TABLE IV

Paired inference for local minus incumbent. Confidence intervals use 10,000 paired bootstrap resamples. The exact sign test excludes numerical ties.

Endpoint	Mean difference	Median difference	95% CI for mean	Better/tie/worse	Sign-test p
Graph cost J	-1.911	-0.719	$[-2.819, -1.060]$	20/0/0	1.91×10^{-6}
Δv (m/s)	-0.232	-0.087	$[-0.338, -0.128]$	20/0/0	1.91×10^{-6}
C^{insp}	0	0	$[0, 0]$	0/20/0	1.0

of observations 4–5 and 7–8, so the geometric change is localized rather than a wholesale replacement of the inspection plan. Figure 6 shows the resulting mission progress. Both routes first exceed the 80% stop target at action 11 and finish at 80.49% inspectable coverage. The reordered route lowers cumulative Δv from 11.955 to 11.902 m/s, a case-specific reduction of 0.0524 m/s (0.44%), and lowers graph cost from 98.126 to 97.691. This illustrative case explains the mechanism; the all-scenario effect estimate remains the evidentiary basis for consistency.

VII. Discussion

The candidate-node state strengthens the formulation beyond static next-best-view selection because covered area, current terminal node, selected observations, and remaining budget enter the decision explicitly. Proposition 2 establishes Markov sufficiency under the fixed-graph assumptions. The reported algorithm does not claim to solve the resulting Bellman recursion on the 48-node pool. It uses the state and edge abstraction to ensure that every reordered transition is evaluated from the correct source and that coverage is credited only after an admissible motion.

The paired result is stronger than a single favorable route but narrower than a general robustness claim. Six-dimensional initial-state perturbations change the first-source transfers and can change the feasible incumbent order, while candidate-to-candidate geometry remains fixed. The outcome-independent case study shows that even two localized order exchanges can alter the cumulative control burden without changing the observation set, action count, or final coverage. The 20/20 direction of improvement and confidence intervals quantify consistency over this designed region. They do not supply a probability of improvement in flight because the Latin-hypercube design is not an operational distribution and does not vary mesh, camera, actuator, or dynamics parameters.

The no-degradation theorem applies only to the shared graph objective and only when the base sequence is feasible on the same finite graph. In the paired data, lower graph cost coincides with lower physical Δv, but the theorem alone would not guarantee that coincidence because the stage cost also contains tracking and action terms. It also provides no improvement guarantee when the incumbent fails the shield. A future method for that case requires either repairing the base sequence or expanding the search beyond reversal moves.

The relation to formation-based observation is similarly complementary. Near-natural relative ellipses and closed-loop formation control around a rotating station can provide lower-effort motion structure and stronger stability analysis [4]. OrbInspect instead resolves which surface regions are visible, how prescribed observations are credited, and which observation should follow the current state. A useful synthesis would generate SOOA candidates from near-natural orbit families, then select among them using the present coverage and camera model. This could reduce transfer effort while preserving explicit target-level inspection accounting.

The safety evidence remains limited to the stated model. HCW translation assumes a circular chief and a fixed LVLH target. Clearance samples and swept surface-intersection tests use the complete deterministic ISS mesh, which removes the previous geometry inconsistency but does not create a closed-loop proof under navigation error, target rotation, delayed commands, actuator uncertainty, or unmodeled disturbances. The paired study disables the optional passive-drift audit, so no passive-safety conclusion should be drawn. The full solver also has no continuous-space optimality guarantee.

The camera evidence has comparable boundaries. The visibility model includes range, field of view, incidence, and mesh occlusion, but the offline graph credits a prescribed terminal observation rather than simulating exposure interruption or closed-loop dwell. Illumination, blur, plume contamination, camera slew, and detailed attitude dynamics are omitted. ROS 2 and Gazebo replay remain implementation work and are not treated as validation in the present manuscript.

A flight-oriented development path should first validate closed-loop replay on Ubuntu 24.04 with ROS 2 Jazzy, since the current ROS section is intentionally a placeholder. Batched full-mesh edge evaluation and cached passive audits can then support a study in which passive drift is enabled. Uncertainty-inflated keep-out volumes, camera-slew and illumination constraints, plume restrictions, navigation error, and a rotating body frame should follow. Higher-fidelity propagation, including an optional Basilisk backend, can finally be introduced while retaining HCW as a transparent regression reference.

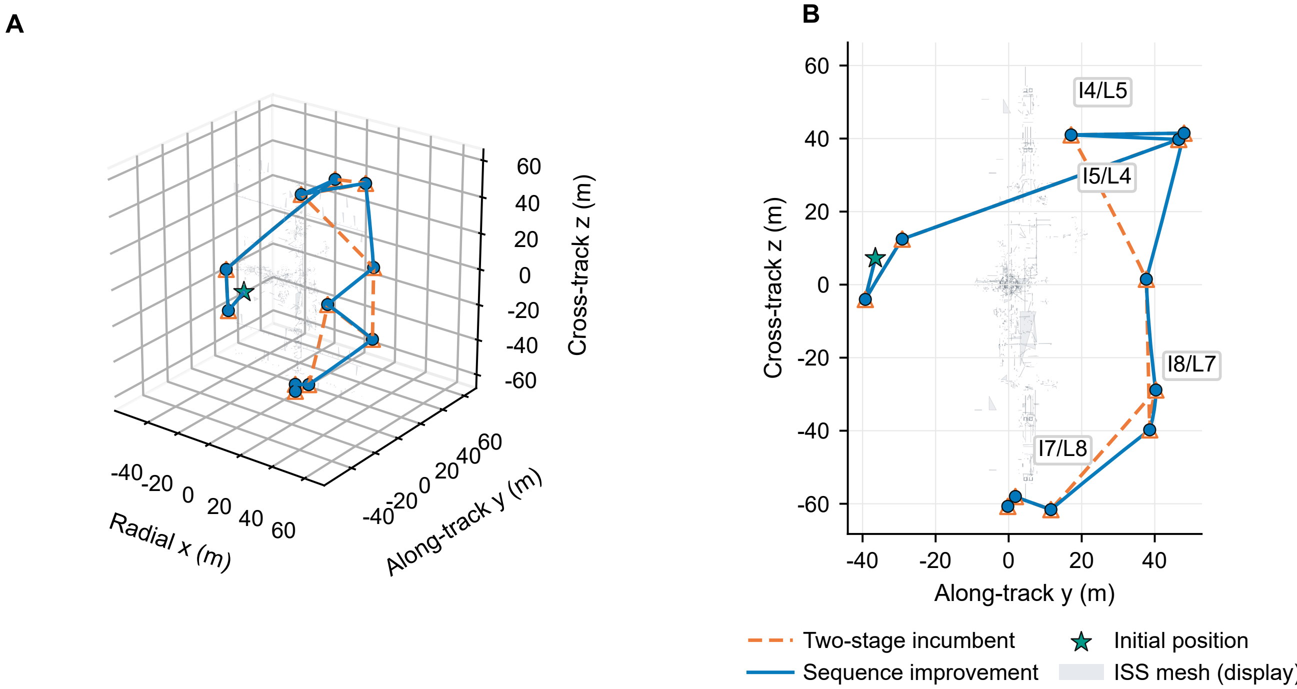

Fig. 5. Outcome-independent trajectory case study for case 9. (a) Complete saved HCW trajectories and terminal observation poses in the LVLH frame. (b) Along-track–cross-track projection; labels $Ik/L\ell$ give the incumbent and local action indices at the four reordered poses. The star is the shared initial position. For legibility, the rendering displays a deterministic sample of 2,400 of the 247,525 ISS triangles; visibility, clearance, and swept-intersection tests use the complete mesh.

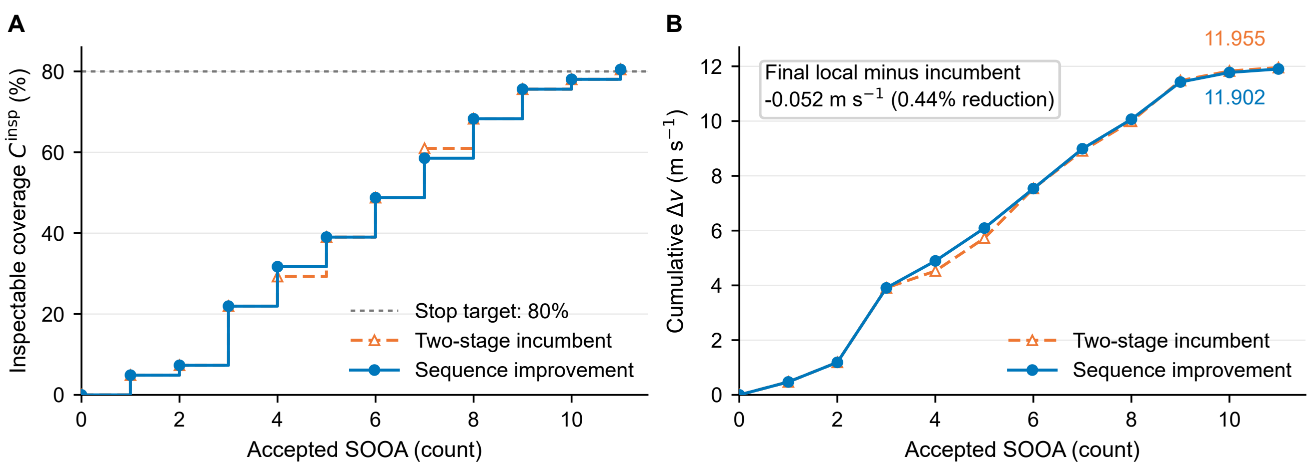

Fig. 6. Mission progress for the design-selected case. (a) Inspectable coverage credited at accepted SOOA endpoints; both routes reach 80.49% after 11 actions. (b) Cumulative Δv; the locally reordered route finishes at 11.902 m/s, compared with 11.955 m/s for the incumbent. These single-case curves are illustrative and do not replace the paired uncertainty analysis in Table IV.

VIII. Conclusion

This paper formulated orbital inspection as safety-shielded sequence improvement over finite-duration observable orbital arcs. The Markov state retains the current candidate node, covered targets, selected nodes, and remaining budget; a deterministic incumbent and fully re-audited reversal moves provide a transparent improvement mechanism. Full-mesh bounding-volume queries now supply visibility, sampled clearance, and swept surface-crossing checks from one consistent ISS representation.

Across 20 predeclared initial-state pairs, both methods were feasible and reached identical 80.49% inspectable coverage. Sequence improvement lowered Δv in every pair by a mean 0.232 m/s, or 1.92%, with a 95% paired-bootstrap interval of 0.128 m/s to 0.338 m/s and exact sign-test $p = 1.91 \times 10^{-6}$. This is evidence for modest, consistent improvement within a fixed sampled graph—not for passive safety, continuous-time robustness, continuous-space optimality, or ROS execution performance. Those claims require the planned Ubuntu/Jazzy closed-loop experiments and higher-fidelity uncertainty studies.

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